

XINGDA LYU

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[Website](#) ◇ [Google Scholar](#) ◇ [LinkedIn](#)

EDUCATION

University of Washington, Seattle

Expected Jun 2027

Dual B.S. degrees in Statistics (Data Science Track) and Informatics (Honors)

Cumulative GPA: 3.9/4.0

Selected coursework: Statistical Machine Learning, Artificial Intelligence, Real Analysis, Data Structures and Algorithms, Mathematical Statistics & Probability I–III, Generalized Linear Models, Databases and Data Modeling.

PUBLICATIONS & MANUSCRIPTS

Kaur, Kirandeep; **Lyu, Xingda**; and Shah, Chirag, 2026. *Position: Knowing Isn't Understanding: Re-grounding Generative Proactivity with Epistemic and Behavioral Insight*.

Accepted at the International Conference on Machine Learning (ICML 2026). [\[paper\]](#)

Lyu, Xingda; Luo, Xinye; Lu, Honglin; and Yang, Shiqi, 2026. *PROS: A Proactive Multimodal Refinement Agent for Scientific Posters*.

Accepted at the Workshop on Human-Interactive Generation and Editing (HiGen), CVPR 2026.

Lyu, Xingda; Lyu, Gongfu; Yan, Zitai; and Jiang, Yuxin, 2025. *P-RAG: Prompt-Enhanced Parametric RAG with LoRA and Selective CoT for Biomedical and Multi-Hop QA*.

In Proceedings of the International Conference on Computing and Data Science. [\[paper\]](#)

SELECTED PRESENTATIONS

Lyu, Xingda, 2026. *Decision-Making Under Unknown Unknowns in Proactive Agentic Systems*.

Lightning talk and poster presentation, ACM CHIIR Workshop on Human-Centered Proactive & Personalized Agents. [\[accepted papers\]](#)

Kaur, Kirandeep; Gupta, Vinayak; **Lyu, Xingda**; Gupta, Aditya, and Shah, Chirag, 2026. *Proactive Agents for Knowledge Gap Navigation*.

Poster presentation, Amazon Research Day 2026, AI Co-Scientist. [\[poster\]](#)

RESEARCH EXPERIENCE

InfoSeeking Lab

Nov 2025 – Present

Supervisor: [Prof. Chirag Shah](#); PhD Mentor: Kirandeep Kaur

University of Washington

- Conduct research on proactive AI agents that support users under uncertainty, incomplete context, and unarticulated information needs.
- Co-authored an ICML 2026 position paper on epistemic and behavioral foundations for proactive agentic systems.
- Developing empirical extensions of this research direction through benchmark design, model-based experiments, and analysis of proactive agent behavior under epistemic uncertainty.

Research Project Lead, PROS

Sep 2025 – Present

Independent Research Project

University of Washington

- Initiated and led PROS, a self-directed research project on proactive multimodal refinement of scientific posters using editable PPTX representations.

- Led a small undergraduate team in building a source-grounded poster-editing pipeline combining structured PPTX representations, LLM/VLM-based diagnosis, and constraint-aware editing.
- Developed an executable taxonomy and staged repair framework for structural validity, layout reallocation, and scientific communication repair, achieving 82.0% macro diagnosis F1 and 73.0% average repair resolution.

Research Project Lead, Algorithms for Big Data Program

Supervisor: *Prof. David P. Woodruff*

Nov 2024 – May 2025

Carnegie Mellon University

- Initiated and led P-RAG, a research project on retrieval-augmented biomedical and multi-hop question answering.
- Co-developed the method by integrating retrieval, LoRA-based parametric adaptation, selective chain-of-thought prompting, and data augmentation.
- Designed and ran experiments on PubMedQA and 2WikiMultihopQA, achieving 93.33% F1 on PubMedQA and improving total score on 2WikiMultihopQA from 17.83 to 33.44.
- Co-first-authored the resulting peer-reviewed conference paper.

PROFESSIONAL EXPERIENCE

Data Science Intern

Amazon

Jun 2025 – Aug 2025

Seattle, WA

- Built automated data-quality pipelines over 300K+ transportation records using Python and SQL.
- Designed routing optimization models projecting \$5M+ in annual savings and 23% route utilization improvement.
- Built anomaly-detection analyses to identify refund gaps and underperforming carriers.

Data Science Intern

Baynovation

Jul 2024 – Sep 2024

San Jose, CA

- Built a refinance decision tool using real-time mortgage rate data.
- Built an interactive dashboard for mortgage scenario analysis and savings visualization.
- Created delinquency risk models using GLM, Random Forest, and XGBoost.

HONORS & SERVICE

UW iSchool Conference Travel Funding Award, to present at ICML

2026

Student Volunteer, International Conference on Machine Learning (ICML)

2026

Co-inventor on three CNIPA-granted utility patents for automated veterinary pharmaceutical production systems

2022–2023

Co-inventor on a Chinese invention patent for a microbial agent for preventing wheat powdery mildew

2022

SKILLS

Languages

Python, R, SQL, Java, JavaScript, HTML/CSS

ML / AI

PyTorch, Hugging Face, scikit-learn, LoRA/PEFT, RAG, LLM fine-tuning, multimodal modeling

Research / Evaluation

benchmark design, ablation studies, statistical modeling, error analysis, qualitative evaluation

Tools

Git/GitHub, Linux, Jupyter Notebook, Flask